

# Milestone 2 Device Configuration Commands

This worksheet documents the final Milestone 2 Packet Tracer configuration. Cisco IOS commands are shown for routers and switches. Packet Tracer server services and server IP settings were configured through the server GUI, not Cisco CLI.

## DLS1: Catalyst 3560 distribution switch

```
enable
configure terminal
hostname DLS1
no ip domain-lookup
ip routing
spanning-tree mode rapid-pvst
spanning-tree vlan 1,10,20,30,40,50,99,100 root primary
vlan 10
    name ADMIN_HR
vlan 20
    name ENGINEERING_DESIGN
vlan 30
    name PRODUCTION_FLOOR
vlan 40
    name FINANCE_PROCURE
vlan 50
    name ICT_TECHNICAL
vlan 99
    name MGMT_SERVERS
vlan 100
    name GUEST_WIFI
vlan 999
    name UNUSED_NATIVE
exit
interface Vlan10
    ip address 192.168.35.33 255.255.255.240
    ip helper-address 192.168.35.114
    no shutdown
interface Vlan20
    ip address 192.168.35.1 255.255.255.224
    ip helper-address 192.168.35.114
    no shutdown
interface Vlan30
    ip address 192.168.35.49 255.255.255.240
    ip helper-address 192.168.35.114
    no shutdown
interface Vlan40
    ip address 192.168.35.97 255.255.255.240
    ip helper-address 192.168.35.114
    no shutdown
interface Vlan50
    ip address 192.168.35.65 255.255.255.240
```

```
ip helper-address 192.168.35.114
no shutdown
interface Vlan99
ip address 192.168.35.113 255.255.255.240
ip helper-address 192.168.35.114
no shutdown
interface Vlan100
ip address 192.168.35.81 255.255.255.240
ip helper-address 192.168.35.114
no shutdown
interface GigabitEthernet0/1
no switchport
ip address 192.168.35.254 255.255.255.252
no shutdown
router ospf 1
router-id 2.2.2.2
passive-interface default
no passive-interface GigabitEthernet0/1
network 192.168.35.0 0.0.0.255 area 0
exit
ip route 0.0.0.0 0.0.0.0 192.168.35.253
interface range FastEthernet0/21 - 24
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
spanning-tree guard root
no shutdown
exit
interface FastEthernet0/21
switchport trunk allowed vlan 1,20,50,99
interface FastEthernet0/22
switchport trunk allowed vlan 1,20,50,99
interface FastEthernet0/23
switchport trunk native vlan 1
switchport trunk allowed vlan 1,30,99,100
interface FastEthernet0/24
switchport trunk native vlan 999
switchport trunk allowed vlan 40,999
interface GigabitEthernet0/2
switchport trunk encapsulation dot1q
switchport mode trunk
switchport nonegotiate
switchport trunk allowed vlan 1,10,99
no shutdown
interface FastEthernet0/2
switchport mode access
switchport access vlan 99
spanning-tree portfast
no shutdown
interface FastEthernet0/3
switchport mode access
switchport access vlan 99
```

```

spanning-tree portfast
no shutdown
end
copy running-config startup-config

```

## R-EDGE: ISR 2911 core router

```

enable
configure terminal
hostname R-EDGE
no ip domain-lookup
interface GigabitEthernet0/0
ip address dhcp
ip nat outside
no shutdown
interface GigabitEthernet0/1
ip address 192.168.35.253 255.255.255.252
ip nat inside
no shutdown
router ospf 1
router-id 1.1.1.1
network 192.168.35.252 0.0.0.3 area 0
default-information originate always
exit
ip route 0.0.0.0 0.0.0.0 dhcp
access-list 1 permit 192.168.35.0 0.0.0.255
ip nat inside source list 1 interface GigabitEthernet0/0 overload
end
copy running-config startup-config

```

## Access switch pattern

Use the matching VLAN and DLS1 uplink from the table below. The end-device port assignments follow the Milestone 1 logical topology.

| Switch  | User VLAN(s) | Access-switch uplink port(s) | DLS1 uplink port(s) | Primary allowed VLANs |
|---------|--------------|------------------------------|---------------------|-----------------------|
| ALS-ADM | 10           | Gi0/1                        | Gig0/2              | 1,10,99               |
| ALS-ENG | 20 and 50    | Gi0/1 and Gi0/2              | Fa0/21 and Fa0/22   | 1,20,50,99            |
| ALS-PRD | 30 and 100   | Gi0/1                        | Fa0/23              | 1,30,99,100           |
| ALS-FIN | 40           | Gi0/1                        | Fa0/24              | 40,999                |

For each access switch, configure the hostname, VLANs, user access ports, uplink, PortFast, and BPDU Guard. Keep unused ports shutdown.

## ALS-ADM

```
enable
configure terminal
hostname ALS-ADM
vlan 10
    name ADMIN_HR
vlan 99
    name MGMT_SERVERS
interface range FastEthernet0/1 - 3
    switchport mode access
    switchport access vlan 10
    spanning-tree portfast
    spanning-tree bpduguard enable
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation restrict
    switchport port-security mac-address sticky
    no shutdown
interface GigabitEthernet0/1
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 1,10,99
    no shutdown
interface Vlan99
    ip address 192.168.35.115 255.255.255.240
    no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config
```

## ALS-ENG

```
enable
configure terminal
hostname ALS-ENG
vlan 20
    name ENGINEERING_DESIGN
vlan 50
    name ICT_TECHNICAL
vlan 99
    name MGMT_SERVERS
interface range FastEthernet0/1 - 3
    switchport mode access
    switchport access vlan 20
    spanning-tree portfast
    spanning-tree bpduguard enable
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation restrict
```

```

    switchport port-security mac-address sticky
    no shutdown
interface range FastEthernet0/17 - 19
    switchport mode access
    switchport access vlan 50
    spanning-tree portfast
    spanning-tree bpduguard enable
    no shutdown
interface GigabitEthernet0/1
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 1,20,50,99
    no shutdown
interface GigabitEthernet0/2
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 1,20,50,99
    no shutdown
interface Vlan99
    ip address 192.168.35.116 255.255.255.240
    no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config

```

## ALS-PRD

```

enable
configure terminal
hostname ALS-PRD
vlan 30
    name PRODUCTION_FLOOR
vlan 100
    name GUEST_WIFI
vlan 99
    name MGMT_SERVERS
interface range FastEthernet0/1 - 3
    switchport mode access
    switchport access vlan 30
    spanning-tree portfast
    spanning-tree bpduguard enable
    switchport port-security
    switchport port-security maximum 1
    switchport port-security violation restrict
    switchport port-security mac-address sticky
    no shutdown
interface GigabitEthernet0/1
    switchport mode trunk
    switchport nonegotiate
    switchport trunk allowed vlan 1,30,99,100
    no shutdown

```

```
interface Vlan99
  ip address 192.168.35.117 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config
```

## ALS-FIN

```
enable
configure terminal
hostname ALS-FIN
vlan 40
  name FINANCE_PROCURE
vlan 99
  name MGMT_SERVERS
vlan 999
  name UNUSED_NATIVE
interface range FastEthernet0/1 - 3
  switchport mode access
  switchport access vlan 40
  spanning-tree portfast
  spanning-tree bpduguard enable
  switchport port-security
  switchport port-security maximum 1
  switchport port-security violation shutdown
  switchport port-security mac-address sticky
  no shutdown
interface GigabitEthernet0/1
  switchport mode trunk
  switchport nonegotiate
  switchport trunk native vlan 999
  switchport trunk allowed vlan 40,999
  no shutdown
interface Vlan99
  ip address 192.168.35.118 255.255.255.240
  no shutdown
ip default-gateway 192.168.35.113
end
copy running-config startup-config
```

## Server and DHCP configuration

### SVR-DC: DHCP + DNS server

SVR-DC is a Packet Tracer Server-PT device, so its IP settings and services were configured through the GUI rather than Cisco IOS CLI.

| Setting | Value |
|---------|-------|
|---------|-------|

|                 |                                                     |
|-----------------|-----------------------------------------------------|
| IPv4 address    | 192.168.35.114                                      |
| Subnet mask     | 255.255.255.240                                     |
| Default gateway | 192.168.35.113                                      |
| DNS server      | 192.168.35.114                                      |
| DHCP service    | On                                                  |
| DNS service     | On                                                  |
| DNS record      | kagiso.local / file-server record to 192.168.35.119 |

DHCP was configured in **SVR-DC -> Services -> DHCP** , not with IOS commands. The seven GUI scopes are:

| Scope              | Network/mask      | Default gateway | DNS server     |
|--------------------|-------------------|-----------------|----------------|
| VLAN10_ADMIN       | 192.168.35.32/28  | 192.168.35.33   | 192.168.35.114 |
| VLAN20_ENGINEERING | 192.168.35.0/27   | 192.168.35.1    | 192.168.35.114 |
| VLAN30_PRODUCTION  | 192.168.35.48/28  | 192.168.35.49   | 192.168.35.114 |
| VLAN40_FINANCE     | 192.168.35.96/28  | 192.168.35.97   | 192.168.35.114 |
| VLAN50_ICT         | 192.168.35.64/28  | 192.168.35.65   | 192.168.35.114 |
| VLAN99_MANAGEMENT  | 192.168.35.112/28 | 192.168.35.113  | 192.168.35.114 |
| VLAN100_GUEST      | 192.168.35.80/28  | 192.168.35.81   | 192.168.35.114 |

Evidence: 05\_svr\_dc\_dhcp.png and 06\_svr\_dc\_dns.png .

### SVR-FP: file/application server on management VLAN

SVR-FP is also configured through the Packet Tracer GUI.

| Setting         | Value           |
|-----------------|-----------------|
| IPv4 address    | 192.168.35.119  |
| Subnet mask     | 255.255.255.240 |
| Default gateway | 192.168.35.113  |
| HTTP service    | On              |

Evidence: 07\_svr\_fp\_http.png .

### Final ACL policy (CR4)

The seven named inbound ACLs below are the final DLS1 policy. IOS sequence numbers are shown in their final corrected order. The file/application server is 192.168.35.119 .

### ACL-ENG-IN

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Engineering traffic cannot reach Finance, but may reach the file server and other permitted destinations.

### **ACL-ADMIN-IN**

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Admin traffic cannot reach Finance, but may reach the file server and other permitted destinations.

### **ACL-FIN-IN**

```
10 deny ip any 192.168.35.0 0.0.0.255
20 permit ip any host 192.168.35.119
30 permit ip any any
```

Finance is isolated from internal address space by the final applied policy.

### **ACL-PRD-IN**

```
10 deny ip any host 192.168.35.119
20 deny ip any 192.168.35.96 0.0.0.15
30 permit ip any any
```

Production cannot reach the file server or Finance.

### **ACL-ICT-IN**

```
10 deny ip any 192.168.35.96 0.0.0.15
20 permit ip any 192.168.35.112 0.0.0.15
30 permit ip any any
```

ICT cannot reach Finance and may reach the management/server subnet.

### **ACL-MGMT-IN**

```
10 permit ip 192.168.35.64 0.0.0.15 any
20 permit ip 192.168.35.112 0.0.0.15 any
30 deny ip any any
```

Only ICT and the management/server subnet are permitted through the management policy.

## ACL-GUEST-IN

```
10 deny ip any 192.168.35.0 0.0.0.255
20 permit ip any any
```

Guest traffic is denied access to internal RFC1918 project networks and permitted elsewhere.

The configured ACL names, entries, and counters are documented in `13_dls1_acls.png`.

## Verification commands

Run these before closing the M2 evidence pack:

```
show vlan brief
show interfaces trunk
show ip interface brief
show ip route
show ip ospf neighbor
show spanning-tree root
show interfaces status
show access-lists
ping 192.168.35.119 source 192.168.35.1
ping 192.168.35.119 source 192.168.35.65
ping 192.168.35.119 source 192.168.35.97
nslookup file-server 192.168.35.114
```

Expected result: authorised Engineering / Admin / ICT users reach the server, while Finance and Production attempts are denied. Capture the successful ping, denied ping, DHCP lease output, and ACL hit counters as the final evidence set.

## Final evidence checklist

- ✓ `show vlan brief` on DLS1 - `08_dls1_vlan_brief.png`
- ✓ VLSM SVI addresses and DHCP helpers - `26_dls1_vlsm_svis_and_helpers.png`
- ✓ `show interfaces trunk` on DLS1 and access switches - `09_dls1_trunks.png`
- ✓ `show ip ospf neighbor` on R-EDGE and DLS1 - `11_dls1_ospf.png`
- ✓ DHCP lease screenshots for VLAN 20 and VLAN 10 - `01_engineering_dhcp.png`, `02_admin_dhcp.png`
- ✓ DHCP lease screenshots for VLAN 100 and VLAN 40 - `08_guest_dhcp.png`, `09_finance_dhcp.png`
- ✓ ALS-ADM VLAN and trunk verification - `25_als_adm_vlan_and_trunk.png`
- ✓ DHCP service screen on SVR-DC (GUI, not `show ip dhcp binding`) - `05_svr_dc_dhcp.png`
- ✓ Successful ping from Engineering to `192.168.35.119` - `03_engineering_server_ping.png`
- ✓ Failed ping from Production to `192.168.35.119` - `04_production_server_ping_denied.png`
- ✓ `show access-lists` with deny counters incrementing - `13_dls1_acls.png`
- ✓ STP root and redundant-link evidence - `15_stp_before_failover_part1.png`, `16_stp_before_failover_part2.png`, `17_stp_before_failover_part3.png`, `18_stp_after_failover_part1.png`, `22_stp_restored_part1.png`

- ☒ Final Packet Tracer save for Milestone2/CLI066\_Mbodi\_40779750\_M2\_CONFIGURED.pkt